

**CD/CO/CS/
IT126**

DATA STRUCTURES

L T P C Int Ext
4 - - 4.0 30 70

Semester II [First Year]

COURSE OBJECTIVES:

The main objectives of this course are to:

1. To illustrate operations of linear and non-linear data structure
2. To demonstrate computational problems using suitable data structures

COURSE OUTCOMES:

After successful completion of the course, the students are able to:

1. Analyze the performance of algorithms to evaluate their efficiency.
2. Apply searching, sorting and hashing techniques to organize and retrieve data efficiently.
3. Develop Linear a structures to manipulate sequential data and solve computational problems.
4. Implement Non-Linear data structures to solve real world problems.

UNIT I

[CO:1] (15)

Introduction: Basic Concepts-Algorithm Specification, Data Abstraction, Performance Analysis-Time complexity, Space complexity, Asymptotic Notations.

Searching and sorting: Linear Search, Binary Search, insertion sort, selection sort. Abstract Data Type, Array ADT, Dynamic memory allocation.

UNIT II

[CO:2] (15)

Lists: Singly Linked Lists, Double linked list, Circular Linked Lists: Operations & their algorithms, Polynomials: Addition, Multiplication.

Hashing: Static Hashing-HashTables, Hashing Functions, Over flow Handling.

UNIT III

[CO:3] (16)

Stacks and Queues: Stack ADT, Queue ADT, Stack applications, Multiple Stacks and Queues, Dynamically Linked Stacks and Queues.

UNIT IV

[CO:4] (14)

Trees: Introduction, binary trees, Binary Tree Traversals, Binary Search Trees, AVL Trees, Heaps, Heap sort, B- Trees and B+ Trees.

Graphs: The Graph Abstract Data Type, representations of graphs, Elementary Graph Operations **DepthFirstSearch, BreadthFirstSearch, ConnectedComponents.**

LEARNING RESOURCES:

TEXT BOOK:

Ellis Horowitz, Sartaj Sahni, Susan Anderson-Freed, "Fundamentals of Data Structures in C", Second Edition, University Press, 2008.

REFERENCE BOOK(s):

1. Mark Allen Weiss, "Data Structures and Algorithm Analysis in C", 2nd Edition, Pearson Education, 1997.
2. Y. Langsam, M.J. Augenstein and A.M. Tenenbaum, Data Structures Using C, Pearson Education Asia, 2004.

3. Aho, Hopcroft and Ullman," Data Structures and Algorithms", Pearson Education, 1983. Jean Paul Trembly and P.G. Sorenson, An Introduction of Data Structures with Applications.